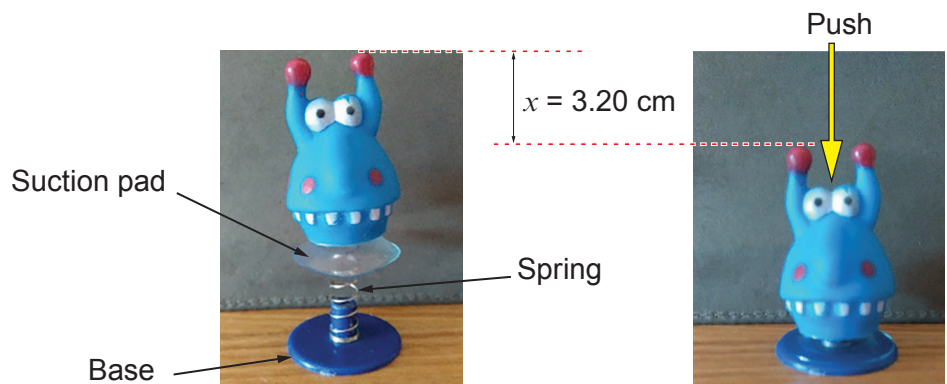


2. A student is asked to find a value for the spring constant, k , of a spring used in a jumping toy. Pushing down on the toy compresses the spring allowing the suction pad to stick to the base. After a few seconds the force on the suction pad decreases and the toy ‘jumps’ into the air.



The student measures the jump height for five separate jumps using a metre rule. The results are shown in the table.

Jump number	1	2	3	4	5
Jump height / cm	48	52	54	46	49

- (a) (i) Calculate the mean jump height, h_{mean} along with the **absolute** uncertainty in its value. [2]

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- (ii) Calculate the **percentage** uncertainty in h_{mean} . [1]

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- (b) (i) The student applies a principle of physics to the jumping toy to show that:

$$k = \frac{2mgh_{\text{mean}}}{x^2}$$

where: m is the mass of the toy and x is the compression of the spring.

State which principle was used and explain how it can be applied to derive the formula shown. [2]

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(ii) The student uses an accurate balance to measure the mass, m , of the toy to be 48.40 g and digital callipers to measure the compression of the spring, x , to be 3.20 cm. He decides **not** to determine the uncertainty in these measurements.

I. Explain why it is reasonable for the student to ignore these uncertainties. [2]

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II. Calculate k along with the **absolute** uncertainty in its value. Give both values to an appropriate number of significant figures. [3]

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III. State whether your answer to (b)(ii)(II) is likely to be smaller than or greater than the actual value for k . Justify your answer. [2]

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(c) Describe one practical procedure by which the student could reduce the uncertainty in k . [1]

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